DS 5230 – Final Project

I. Introduction

1. What did you do? Why did you do it?

This project involved developing and comparing time-series machine learning models to predict future product demand for a Philippine e-commerce business. The primary goal was to address the "dual challenge" of inventory management: preventing stock outs of high-demand items while avoiding the accumulation of excess, capital-trapping inventory for low-demand products. By transitioning from reactive to data-driven strategies, the business can optimize supply chains, reduce waste, and improve profitability.

II. Background

The field of demand forecasting has evolved from localized statistical methods to global, deep learning architectures. Historically, the ARIMA model has served as the industry standard for capturing linear relationships and seasonal patterns, though it often struggles with non-linear market shocks. More recently, Global Forecasting Models like DeepAR have emerged, utilizing recurrent neural networks to learn correlations across an entire product catalog, which is particularly useful for new items with limited history. However, research suggests that the choice of algorithm must align with the specific characteristics of the data; while deep learning thrives on massive datasets, simpler statistical or feature-engineered models often remain competitive in smaller-scale retail environments. This project builds on these insights by evaluating which approach provides the most tangible business value for a medium-sized product catalog.

III. Methods/Analysis

1. Basic demographics of your data

The study utilized 74,802 transactional records from a small Southeast Asian e-commerce business.

Scope: 37 months (January 2023 – January 2026).

Scale: 773 distinct products across 8 categories.

Key Characteristics: The data is 62.2% sparse (many products have no sales in a given month) and the target variable is heavily right-skewed, with most products selling only 1–10 units per month.

2. How did you format the data?

Transaction-level data was aggregated into monthly product-level observations. Input feature vectors were enriched with:

Lag Features: Sales from the previous 1, 2, and 3 months.

Rolling Features: A 3-month rolling average.

Metadata: Product ID, price statistics, growth rates, and encoded category.

3. How did you clean the data?

Preprocessing involved dropping six null Item IDs and correcting nine date anomalies. Lag and rolling features with missing values in early months were filled with zero, and infinite values in price change percentages were also replaced with zero.

4. What tests/algorithms did you use?

The study utilized a progression of statistical and machine learning tests to determine the most effective forecasting method for the dataset.

ARIMA (AutoRegressive Integrated Moving Average): Used as a classical statistical benchmark to identify linear patterns and seasonal stability. For products with at least 18 months of history, individual ARIMA (1,1,1) models were fitted using Maximum Likelihood Estimation (MLE).

Simple Moving Average (SMA): Applied to products with 3–17 months of history to provide a rolling average forecast based on the previous three months.

Naive Persistence: Used for products with less than 3 months of history, where the forecast simply equals the value of the previous month.

Gradient Boosting Regressor: A tree-based ensemble model used to evaluate if feature-engineered machine learning could outperform classical methods. It was trained using squared error loss on 16 engineered features.

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Feature Importance Analysis: A test used specifically on the Gradient Boosting model to rank which input variables (e.g., rolling averages, growth rates, lags) contributed most to the predictive power.

IV. Results

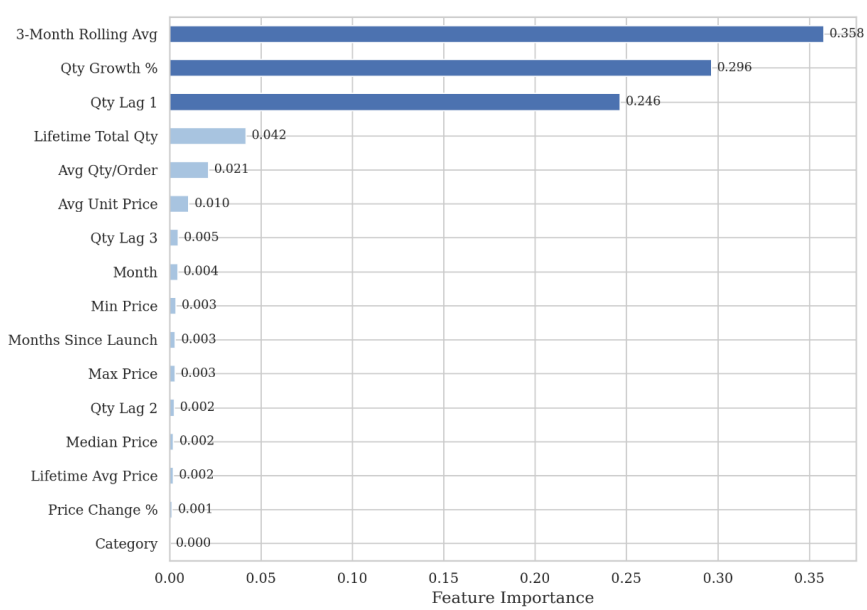
1. Raw results

Gradient Boosting significantly outperformed all other models across every metric.

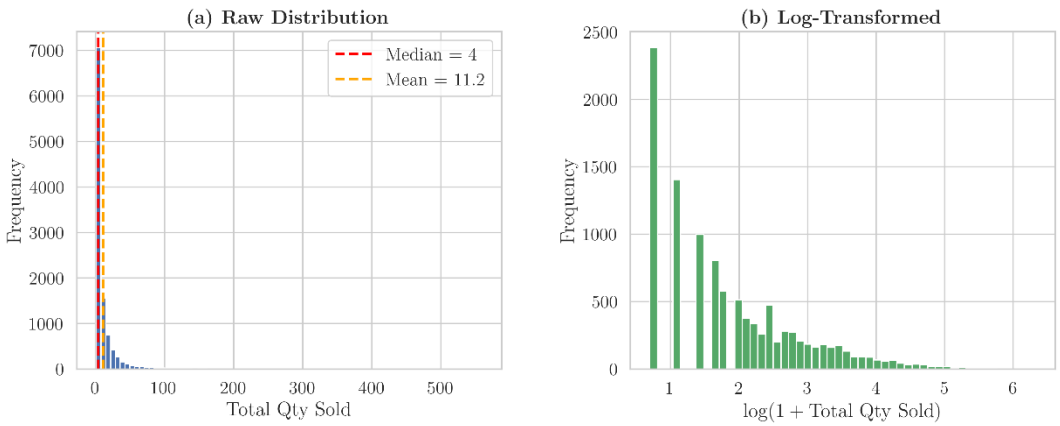
Model	MAE	MAPE	R^2
ARIMA	8.20	110.1%	0.195
DeepAR	8.71	94.4%	0.093
Gradient Boosting	**1.22**	**15.7%**	**0.956**

2. Tables and graphs

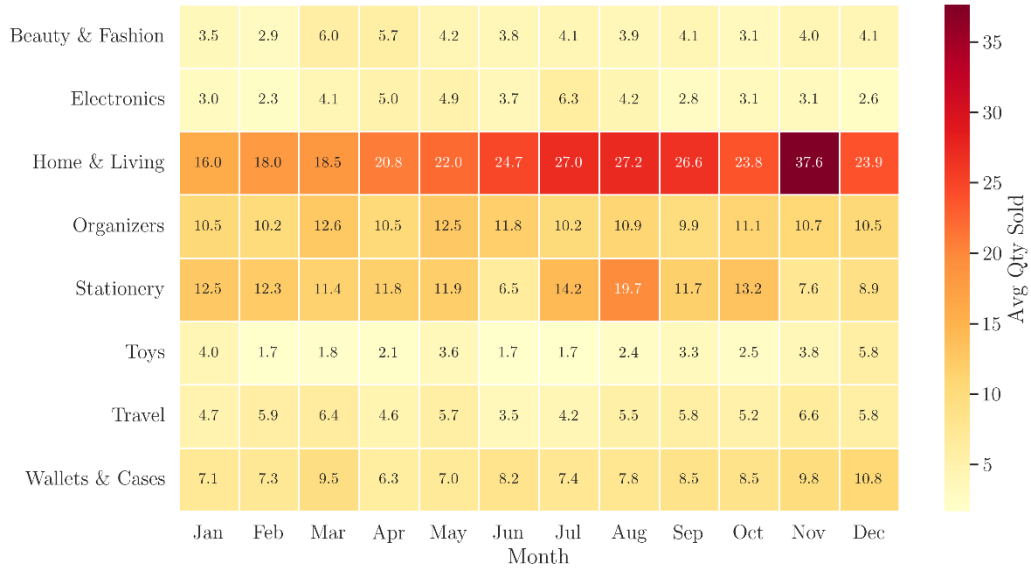
The project included a feature importance bar chart showing that three features (3-month rolling average, quantity growth %, and 1-month lag) account for 90% of the predictive power.



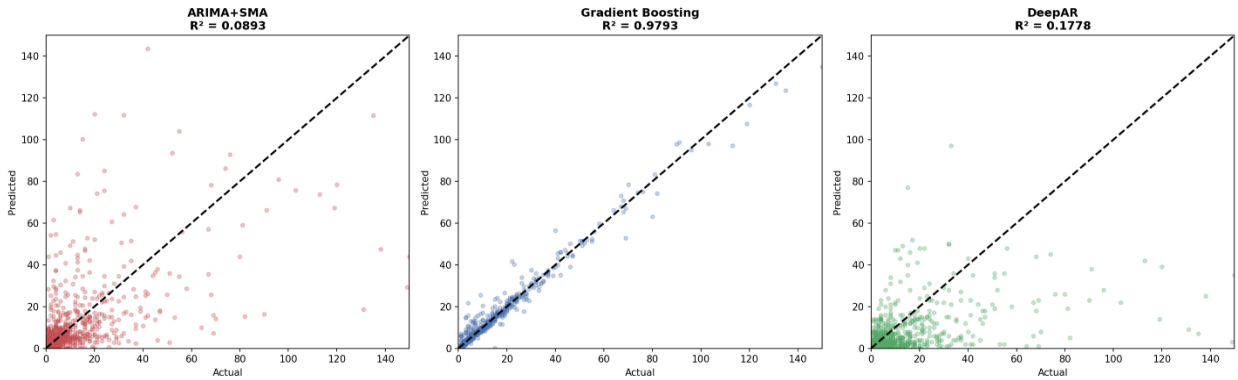
Distribution: The target variable is heavily skewed; log-transformation was used to visualize the density more clearly.

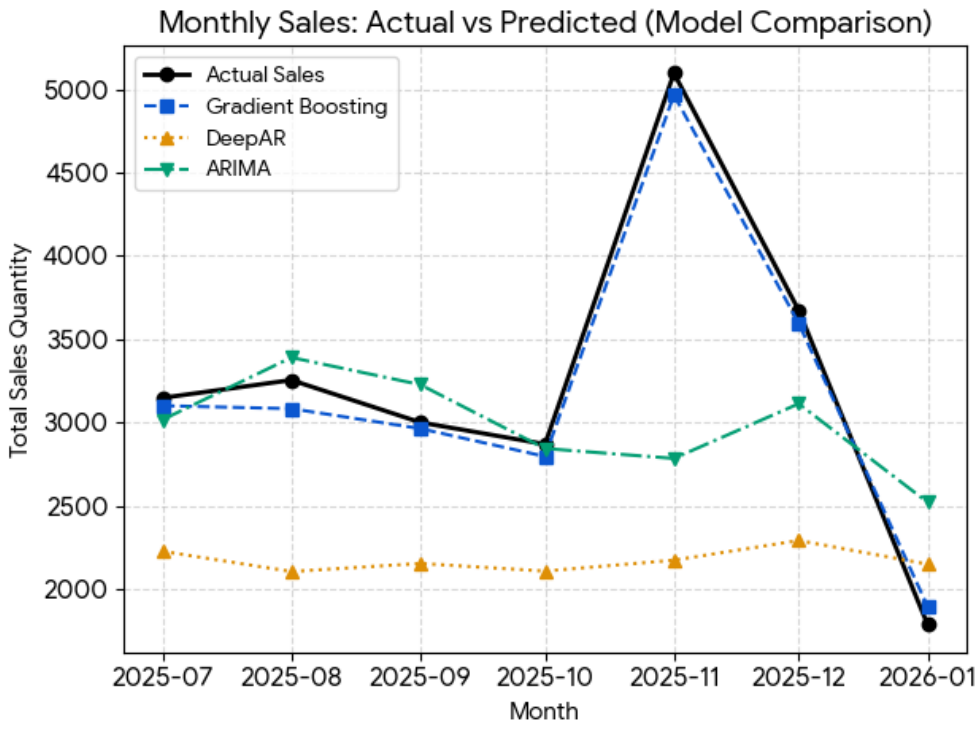


Seasonality: A heatmap shows category-dependent peaks, such as "Home & Living" spiking in November and "Stationery" in August.



Prediction Quality: Scatter plots show Gradient Boosting predictions following the perfect prediction line closely, whereas ARIMA and DeepAR show high variance and scatter.





V. Discussion

1. What do your results mean?

The success of Gradient Boosting (R^2 of 0.956) over DeepAR (R^2 of 0.093) indicates that for smaller datasets with short histories and high sparsity, explicit feature engineering is superior to end-to-end deep learning. DeepAR struggled because 37 months of data is insufficient for an LSTM to learn complex temporal representations.

2. How can your results be used?

Retailers can use these forecasts to set optimal reorder points. Because demand is driven largely by recent momentum (90% predictive power from lag features), businesses should focus on 3-month rolling averages and growth rates to guide their inventory purchases.

VI. Limitations

1. What issues did you run into?

Data Sparsity: 62.2% of the product-month matrix was empty, which disproportionately harmed the performance of sequential models like DeepAR.

Isolation: ARIMA failed because it treats each product as an independent series and could not learn shared seasonal patterns across the catalog.

2. What data is missing?

The models lacked external covariates such as promotional schedules, holidays, competitor pricing, and macroeconomic indicators, which could have improved accuracy.

VIII. Future Work

1. Suggestions for the client

Prioritize Feature Engineering: Focus on tree-based models (like Gradient Boosting) rather than complex deep learning if the product catalog remains medium-sized.

Incorporate External Data: Start tracking promotional events and holidays to better capture demand shocks that lag features cannot predict.

Human Oversight: Treat the model as a decision-support tool; inventory managers should have the final say, especially for new product launches where historical data is missing.

Hybrid Modeling: Explore combining Gradient Boosting point estimates with DeepAR's probabilistic intervals to capture the strengths of both.